V4.2 Toy Model 4H
Relational Action to Metric Response

Rev A - Exploratory Gravity / Metric Bridge Report

Date: 2026.05.26 | *Continuation after Toy Model 4G*

Status: Toy Model 4H derives the graph Poisson response from a native relational action functional, then uses the same trace-reversed metric interface from 4G to test the forward curvature balance. It does not derive the full Einstein-Hilbert action, full Einstein Field Equations, physical gravity, dark matter, dark energy, or physical 3T.

1. Purpose

Toy Model 4G showed that the forward pipeline works if a trace-reversed Poisson-like metric response is supplied. Toy Model 4H asks which part of that rule can be derived from a native graph principle rather than imported.

The result is deliberately split into two layers: the graph equilibrium law and the metric-interface law.

2. Critical correction before interpretation

4H does not derive the whole gravitational law. It derives the scalar graph Poisson response from a relational action. The later mapping from that scalar response into trace-reversed metric perturbation is still an interface ansatz inherited from 4G.

3. Relational action functional

Define a graph response field Phi over the nodes. The native relational action is:

```
A[Phi] = integral [ 1/2 |grad Phi|^2 - S Phi ] dA
```

Meaning of the terms:

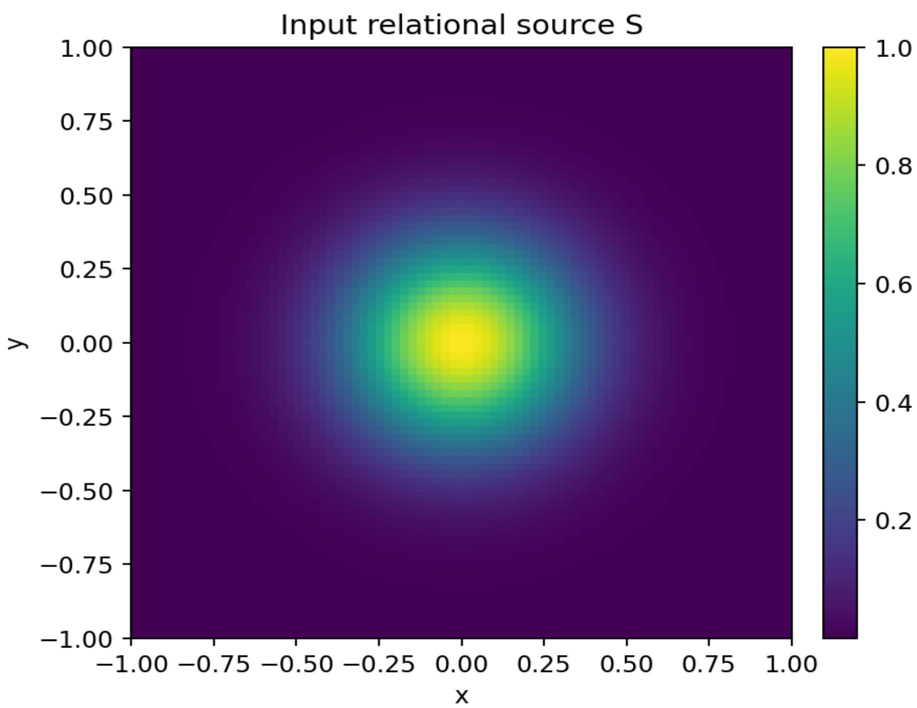
• Phi is the network response field.

• S is the input relational source.

• 1/2 |grad Phi|^2 penalizes uneven response between neighboring nodes.

• -S Phi rewards placing response where the source exists.

• The equilibrium is the field Phi that balances smoothness against source coupling.



4. Variation of the action

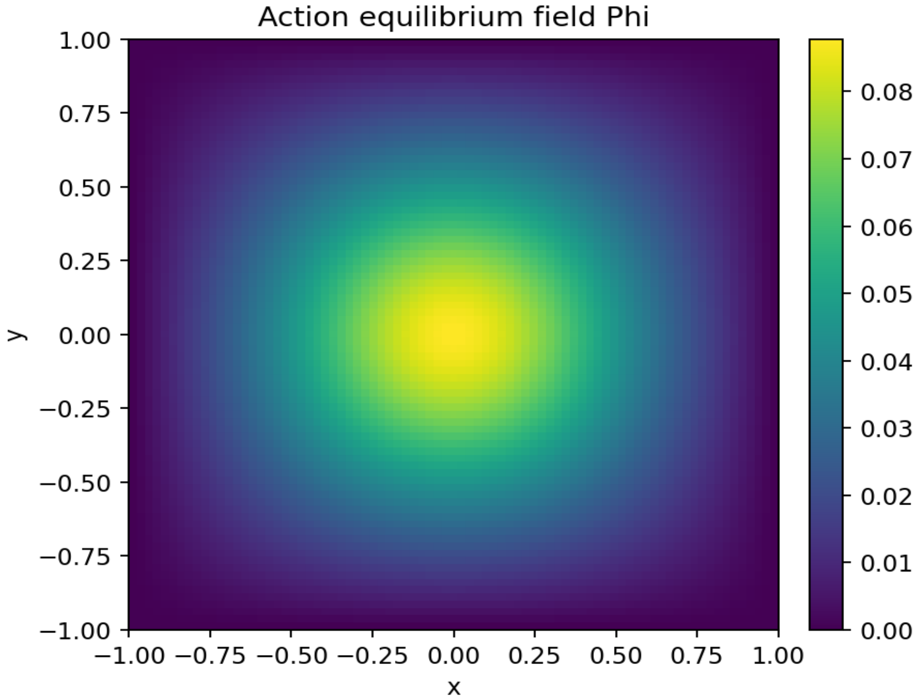
Varying the action with respect to Phi gives the Euler-Lagrange equation:

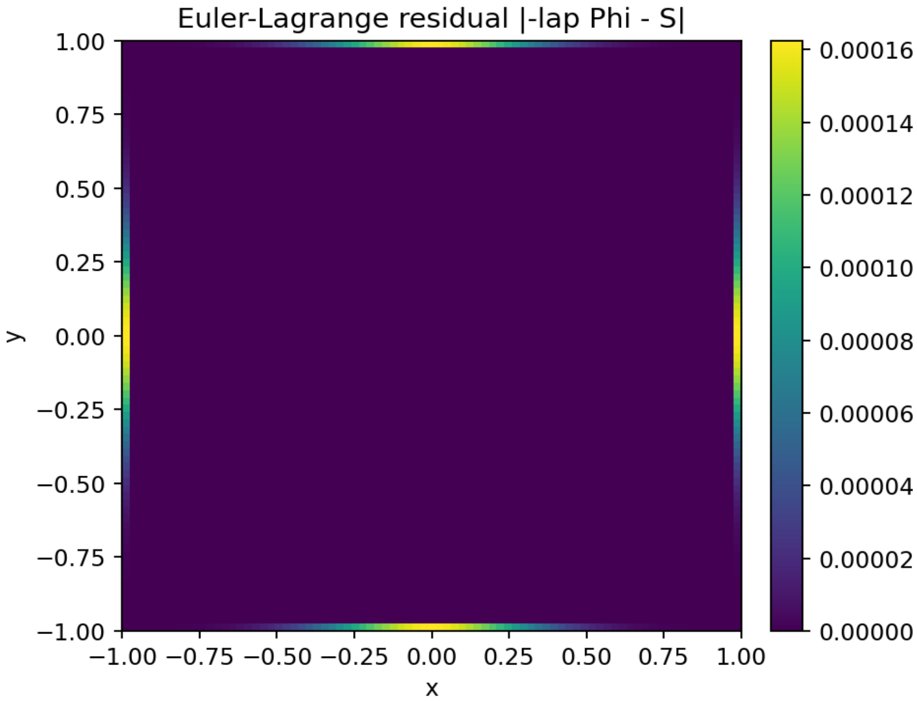
```
delta A / delta Phi = 0
```

```
=> -nabla^2 Phi - S = 0

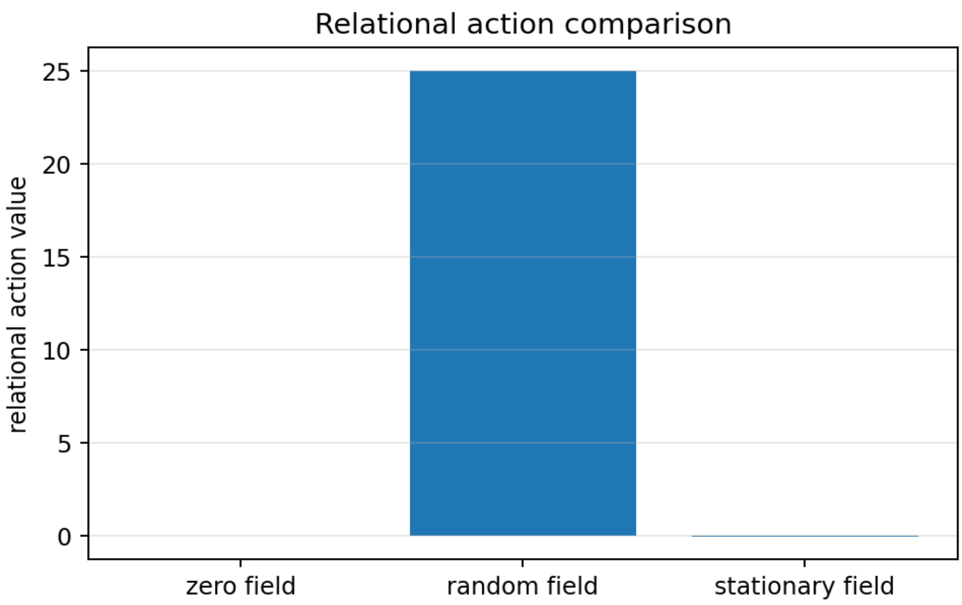
=> nabla^2 Phi = -S
```

Plain meaning: the graph bends its response field only as much as necessary to balance the source while keeping neighboring nodes smooth. This derives the Poisson response used in 4G from an optimization principle.





5. Action diagnostics



Action / variation diagnostic	Value
Action at zero field	0.000000e+00
Action at random field	2.503601e+01
Action at stationary field	-1.283720e-02
Max Euler residual |-lap Phi - S|	3.401723e-13
RMS Euler residual	3.627824e-14
Directional derivative at solution	-1.061467e-02
Directional derivative at zero field	8.625039e-03

The stationary field has the expected near-zero variation residual. This is the native graph-equilibrium result of 4H.

6. Metric interface ansatz

After the action-derived response field is obtained, 4H uses the same weak-field metric interface as 4G:

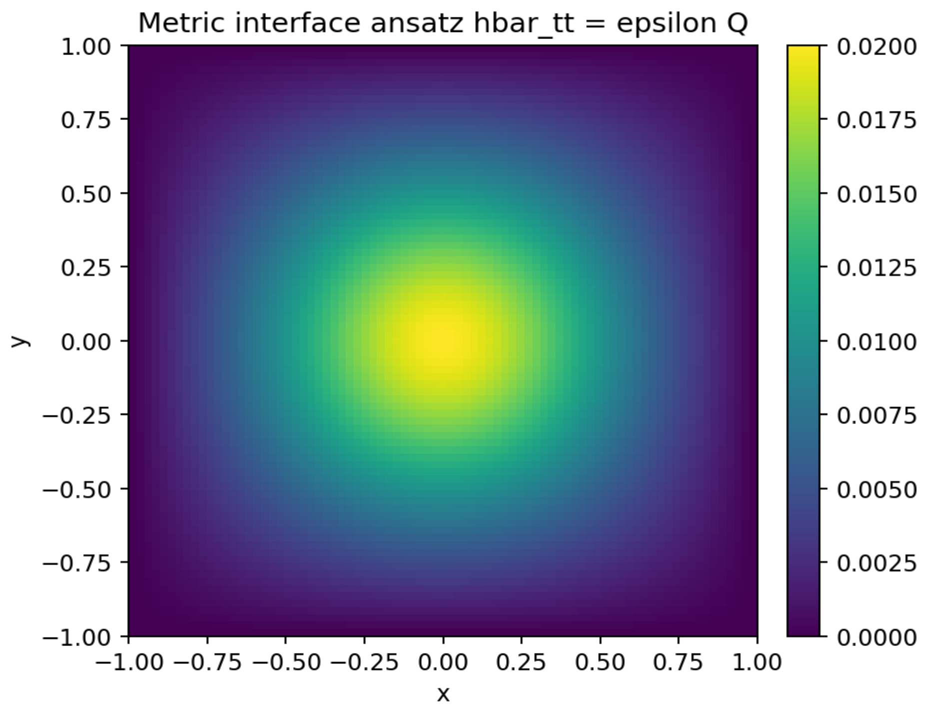
```
Q = Phi / max|Phi|

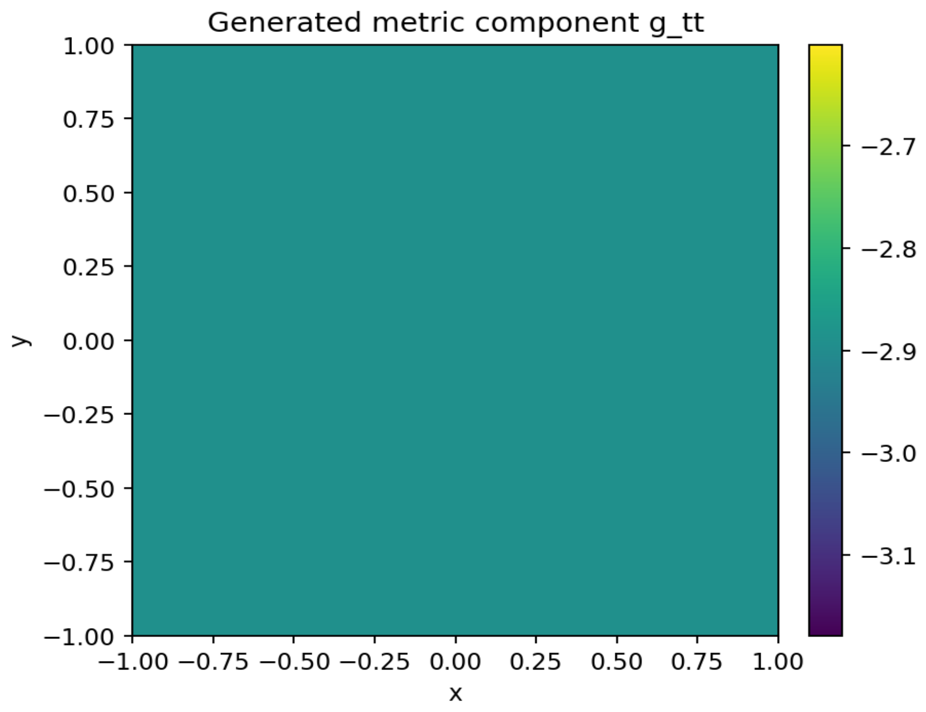
hbar_tt = epsilon Q

h_mu_nu = hbar_mu_nu - eta_mu_nu hbar_trace/(D-2)

g_mu_nu = eta_mu_nu + h_mu_nu
```

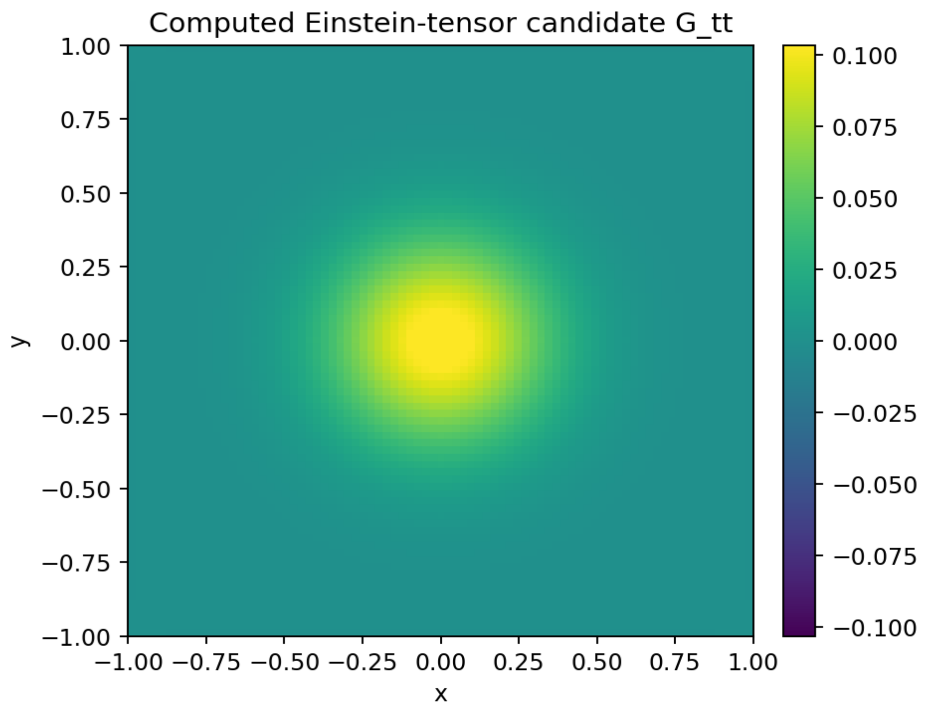
This is the key remaining guardrail. The Poisson response was derived from the graph action. The trace-reversed metric interface was not derived from first principles here.

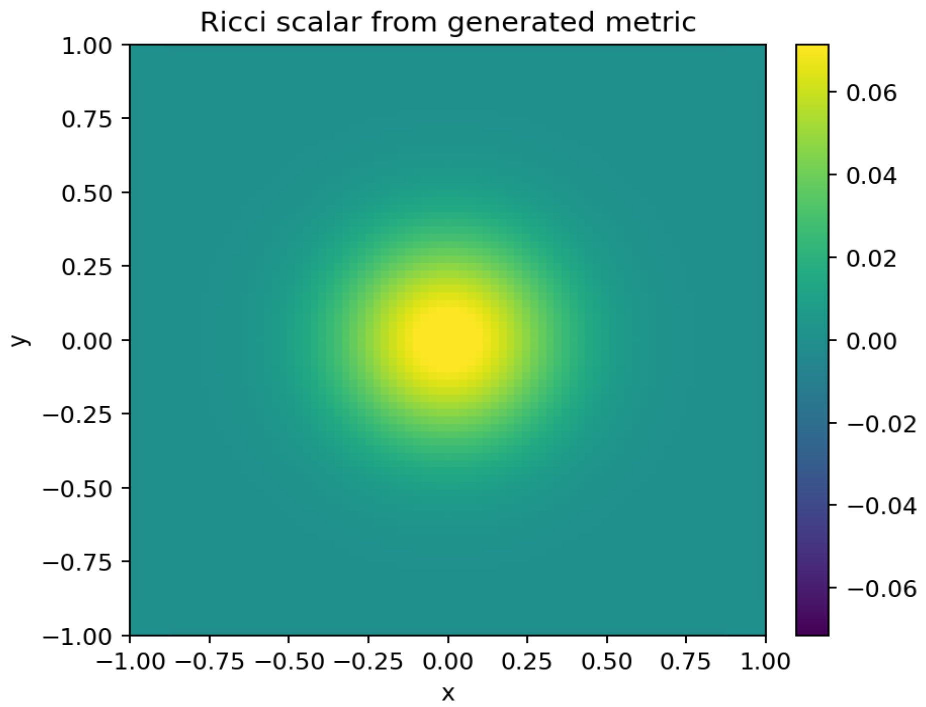




7. Curvature and balance from the generated metric

The generated metric is passed through the same curvature pipeline: Christoffel symbols, Ricci tensor, Ricci scalar, and Einstein-tensor candidate.



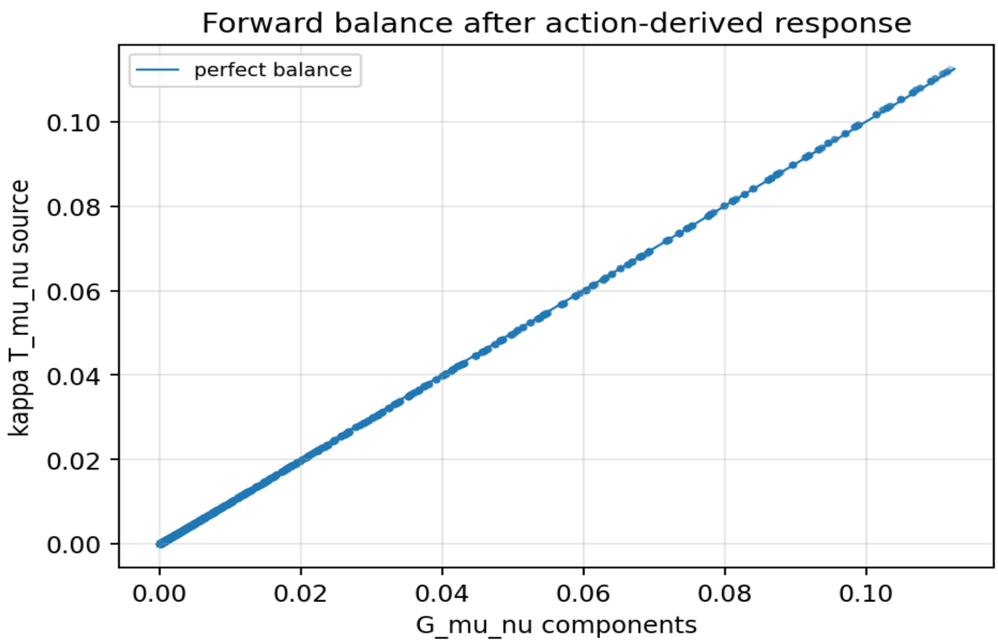


8. Balance test

The field-equation-like balance test is:

```
G_mu_nu ~= kappa T_mu_nu
```

The input source tensor is still minimal and dust-like: T_tt=S, with other components zero.



Balance diagnostic	Value
Fitted kappa, full tensor	1.125261e-01
Full tensor residual	5.534844e-03
Full tensor correlation	0.999985
Fitted kappa, G_tt only	1.125261e-01
G_tt residual	5.534844e-03
G_tt correlation	0.999990

9. Component-by-component check

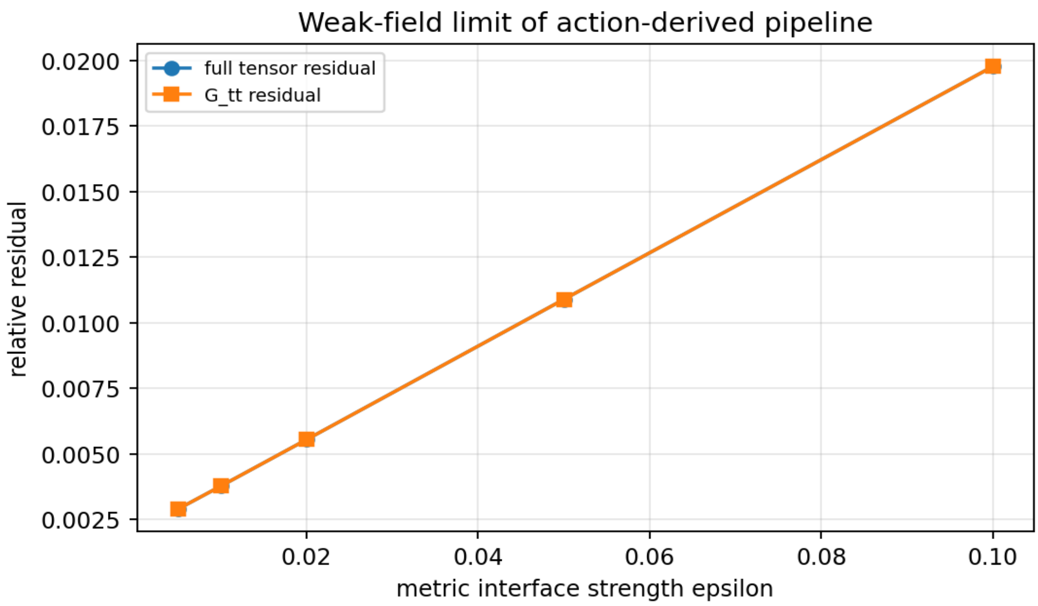
Component	Residual	Correlation	Max |G|	Max |kT|
tt	5.534844e-03	0.999990	1.119311e-01	1.125261e-01
tx	0.000000e+00	1.000000	0.000000e+00	0.000000e+00
ty	0.000000e+00	1.000000	0.000000e+00	0.000000e+00
xx	4.360913e-02	1.000000	6.938894e-18	0.000000e+00
xy	1.000000e+00	0.000000	5.342850e-08	0.000000e+00
yy	4.363473e-02	1.000000	6.938894e-18	0.000000e+00

10. Conservation / Bianchi diagnostics

Conservation diagnostic	Value
Max |nabla_mu T^mu_nu|	0.000000e+00
Mean |nabla_mu T^mu_nu|	0.000000e+00
Max |nabla_mu G^mu_nu|	4.722339e-07
Mean |nabla_mu G^mu_nu|	1.654471e-07

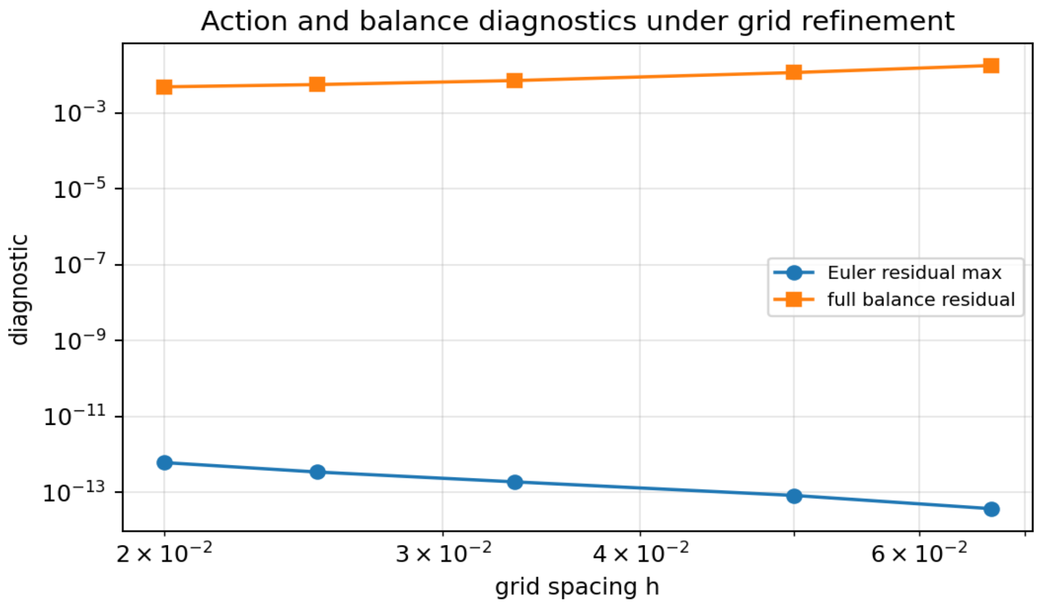
11. Weak-field behavior

The action-derived response was tested at different metric-interface strengths. The residual remains small in the weak-field regime and worsens as the perturbation grows, which confirms this is a linearized toy law.



epsilon	full residual	full corr.	G_tt residual	G_tt corr.	Lorentz sig.
0.005	2.881502e-03	0.999996	2.881502e-03	0.999997	4761/4761
0.010	3.759728e-03	0.999993	3.759728e-03	0.999995	4761/4761
0.020	5.534844e-03	0.999985	5.534844e-03	0.999990	4761/4761
0.050	1.088528e-02	0.999942	1.088528e-02	0.999963	4761/4761
0.100	1.976698e-02	0.999808	1.976698e-02	0.999882	4761/4761

12. Grid refinement



N	h	Euler max	full residual	G_tt residual	max divG	Lorentz sig.
31	0.06667	3.675e-14	1.763935e-02	1.763935e-02	3.089573e-06	529/529
41	0.05000	8.216e-14	1.148196e-02	1.148196e-02	1.813294e-06	1089/1089
61	0.03333	1.870e-13	7.078366e-03	7.078366e-03	8.306858e-07	2809/2809
81	0.02500	3.402e-13	5.545654e-03	5.545654e-03	4.722339e-07	5041/5041
101	0.02000	6.039e-13	4.834121e-03	4.834121e-03	3.037138e-07	7569/7569

13. Core numerical record

Diagnostic	Value
Grid	81 x 81
Interior points	4761

Diagnostic	Value
Lorentzian signatures	4761/4761
Source center	1.000000e+00
Phi center	8.779152e-02
Q center	1.000000e+00
hbar_tt center	2.000000e-02
g_tt center	-2.890000e+00
G_tt center	1.119311e-01
Ricci scalar center	7.746094e-02

14. Interpretation inside the V4.2 framework

Toy Model 4H supports this limited ladder:

```
native graph action A[Phi]
      -> variation delta A = 0
      -> graph Poisson response nabla^2 Phi = -S
      -> metric interface ansatz hbar_tt = epsilon Q
      -> generated metric g_mu_nu
      -> curvature
      -> weak-field balance G_mu_nu ~= kappa T_mu_nu
```

It does not yet support this stronger ladder:

```
3T relational action
      -> trace-reversed metric interface
      -> full tensor metric law
      -> nonlinear Einstein Field Equations
      -> physical gravity / dark sector solution
```

15. What Toy Model 4H achieves

• It derives the graph Poisson response from a native relational action functional.

• It verifies the Euler-Lagrange residual numerically.

• It keeps the forward pipeline from 4G: source -> response -> metric -> curvature -> balance.

• It maintains strong weak-field balance after the action-derived response is used.

• It identifies exactly what remains imported: the trace-reversed metric interface.

16. What Toy Model 4H does not achieve

• It does not derive the trace-reversed metric interface from first principles.

• It does not derive the Einstein-Hilbert action.

• It does not derive nonlinear Einstein Field Equations.

• It does not derive physical stress-energy or Newton's constant.

• It does not solve gravity, dark matter, or dark energy.

• It does not prove physical 3T.

17. Close-out statement for 4H

Toy Model 4H partially derives the constitutive law used in 4G. The scalar graph response is no longer inserted by hand; it follows from minimizing a relational action that balances smoothness cost against source coupling. This yields the Poisson response equation used in the forward pipeline. However, the metric interface step - especially the trace-reversed mapping from scalar response to hbar_mu_nu - remains an ansatz. The correct conclusion is therefore: the graph can derive its own scalar response law, but the tensorial metric-response law is not yet derived.

18. Recommended next chapter

The next logical chapter is Toy Model 4I: Tensorial Relational Action and Metric-Interface Derivation.

1. Upgrade Phi from a scalar response field to a tensorial response hbar_mu_nu.

2. Define a graph action over tensor-valued edge/metric variables.

3. Vary the action with respect to hbar_mu_nu, not only Phi.

4. Check whether trace-reversal or gauge constraints emerge from the variation.
5. Only then discuss an Einstein-equation analogue.

19. Rebind prompt for continuation

```
We are resuming after Toy Model 4H.

4G:
Used source -> graph Poisson response -> trace-reversed metric response -> curvature.
It worked well, but the response law was imported.

4H:
Derived the scalar graph response from a native relational action:

A[Phi] = integral [1/2 |grad Phi|^2 - S Phi] dA

Variation:
delta A / delta Phi = 0
=> -nabla^2 Phi - S = 0
=> nabla^2 Phi = -S

Result:
The graph Poisson response is now action-derived, not inserted by hand.

Then:
Q = Phi/max|Phi|
hbar_tt = epsilon Q
h_mu_nu = hbar_mu_nu - eta_mu_nu hbar_trace/(D-2)
g_mu_nu = eta_mu_nu + h_mu_nu

Balance remained strong in the weak-field regime:
G_mu_nu ~= kappa T_mu_nu

Critical guardrail:
Only the scalar response law was derived.
The trace-reversed tensor metric interface remains an ansatz.

Next recommended chapter:
Toy Model 4I - Tensorial Relational Action and Metric-Interface Derivation.
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